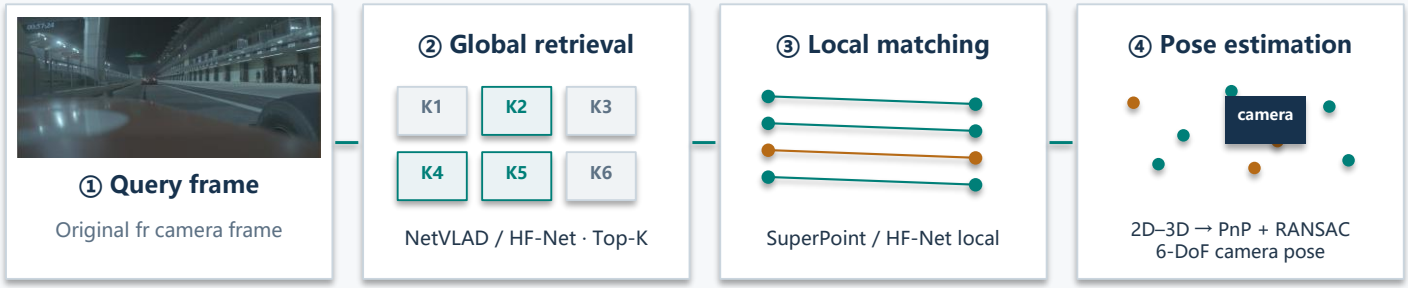
Camera Stream Localization: From Images to Vehicle Pose

HF-Net hierarchical localization (Sarlin et al., arXiv:1812.03506) and independent monocular SfM maps

Goal: estimate base_link pose (x, y, ψ) in map using only on-board JPEG camera streams (~10 Hz).

Paper principle | Coarse-to-Fine localization

HF-Net jointly predicts global and local features: retrieve first, then estimate pose geometrically.



Why hierarchical? Full-database local matching is slow and ambiguous. Top-K retrieval narrows candidates; mutual NN, ratio test, and PnP-RANSAC verify pose. Boxes and lines are schematic.

Why not stereo?

Use one monocular SfM map per camera; fuse poses later.

Calibration	fl / fr / r / rl are more reliable, but intrinsics, orientation, and focal length still differ.
Weak geometry	Baseline and extrinsic accuracy are insufficient for stable stereo depth; errors would affect stereo VO.
Uneven FOV	fl and fr overlap, but fl has a narrower view and fewer features, weakening a stereo pair.

Independent camera poses → downstream fusion.

Offline mapping | Lap2 + Lap4, separate SfM per camera

Images + GT → s-neighbor matching → triangulation → 3D landmarks. Lap3 is held out for validation.

map_fr | **map_r** | **map_rl**

Online localization | image → camera map → camera pose

Default sp_nv_nn: SuperPoint + NetVLAD → Top-5 → mutual NN + ratio test → 2D–3D PnP/RANSAC → camera-to-base_link transform.

Then fuse camera poses in SE(2) / EKF (slides 2 and 4).

Camera Selection, Mapping Strategy, and 3-Camera Fusion

Candidates: fl / fr / r / rl. Compare Lap2-only and s-ordered Lap2+4 maps on the same Lap3 validation queries.

Camera selection

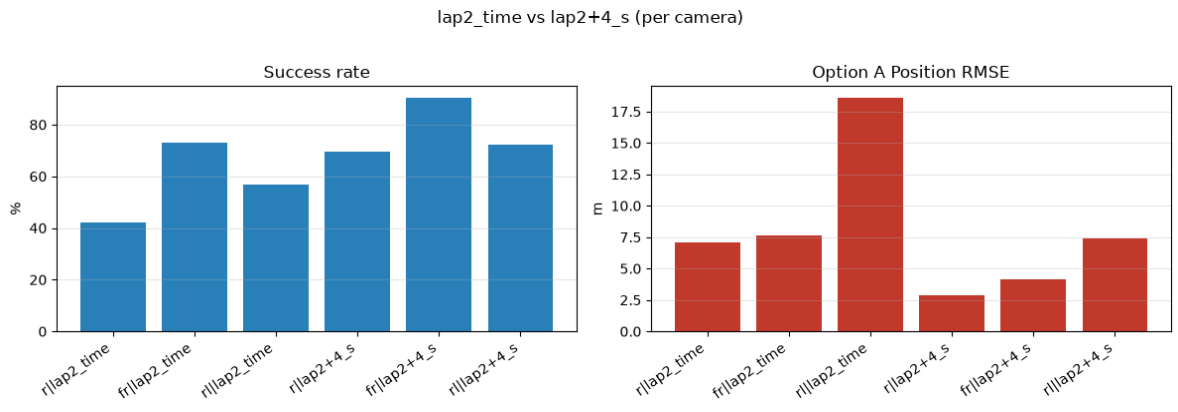
Use fr, r, and rl; exclude fl from the 3-camera run.

① Calibrated set: fl, fr, r, rl.
② fl overlaps fr but yields fewer keypoints.
③ Use r+fr+rl for accuracy and fallback.

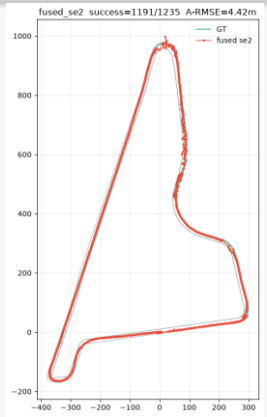
Complementarity in the notebook: when r fails, fr still succeeds on 77.8% of those frames and rl on 48.1%. Select for both accuracy and coverage.

Mapping strategy | Lap3 A-RMSE and success

Neighbors follow racing-line s; RMSE uses valid poses, success = valid / all queries.



Fused trajectory | Lap3, map=Lap2+4



Each camera yields $\mathbf{z}_c = (x_c, y_c, \psi_c)$, PnP inliers $n_{\text{inl},c}$, and optional mean reprojection error $\bar{e}_c$.

Weight: $\omega_c = \mathbf{1}_{\text{ok},c}\, n_{\text{inl},c}\, e^{-\bar{e}_c/(4\,\text{px})}$.

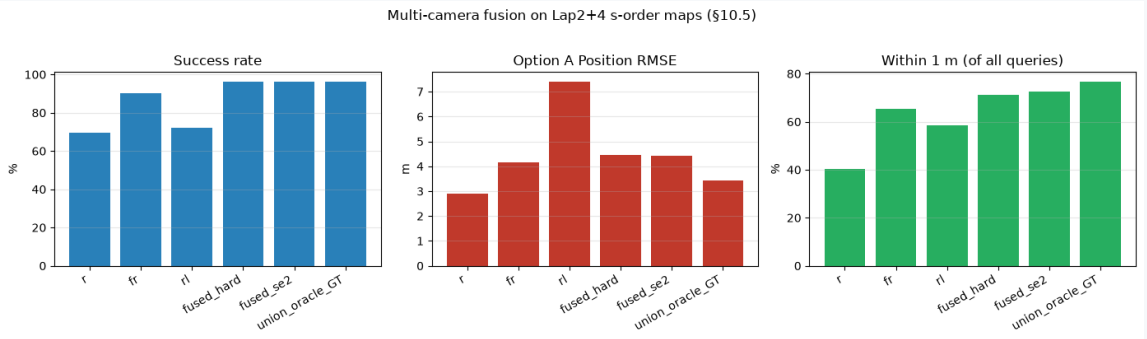
If reprojection error is unavailable, use inliers only.

Anchor at highest weight; reject $\Delta_{xy} > 5\,\text{m}$ or $|\Delta\psi| > 15^\circ$. Average x/y by weight and yaw using atan2 of weighted sin/cos.

Hard mode keeps only the anchor pose.

Single camera vs. fusion | Lap3

fused_se2: 96.4% success, 4.42 m RMSE, 72.7% ≤1 m (all). GT oracle: ideal limit.



Source: 3d-map-test/sfm_hfnet_r_lap2_lap4_s_order_lap3_loc.ipynb (map metrics, fusion metrics, trajectories).

Motion and Visual Errors Along Track Distance s

Odometry provides fast predictions but drifts over a lap. Estimate visual observation noise by track location for EKF updates.

How is position predicted?

Inputs: SE body velocity, acceleration, and yaw rate; SE x/y are not runtime position observations.

$$\Delta \mathbf{p}_b = \mathbf{v}_b \Delta t + \tfrac{1}{2}\mathbf{a}_b \Delta t^2, \quad \psi^- = \mathrm{wrap}(\psi + \omega_z \Delta t).$$

Rotate by mid-step yaw; reset velocity at each SE sample to avoid double integration. The first pose initializes state.



Notebook results:
v-only xy median / p95: 8.37 / 22.9 m; v+a: 8.26 / 23.0 m. Both end 15.7 m off.

Acceleration does not remove lap-scale drift; vision must correct it.

Motion noise | Robust σ via MAD

Step residuals between SE measurements and finite-difference references.

Channel	MADσ	Unit
σ_{v_x}	0.073	m/s
σ_{v_y}	0.020	m/s
σ_{a_x}	10.59	m/s²
σ_{a_y}	8.63	m/s²
σ_{ω_z}	0.0031	rad/s

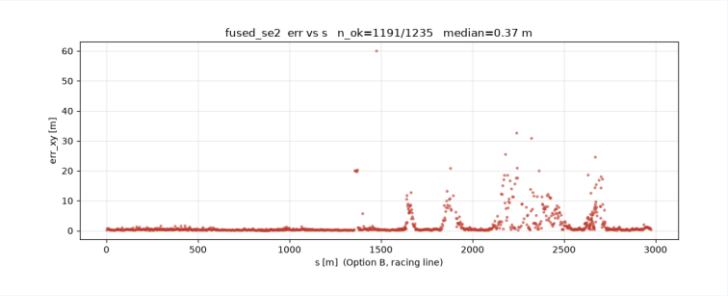
$$\mathbf{v}_{\mathrm{ref}} = R(\psi)^{\mathrm{T}} \Delta \mathbf{p}/\Delta t, \quad \mathbf{a}_{\mathrm{ref}} = \Delta \mathbf{v}_{\mathrm{ref}}/\Delta t.$$

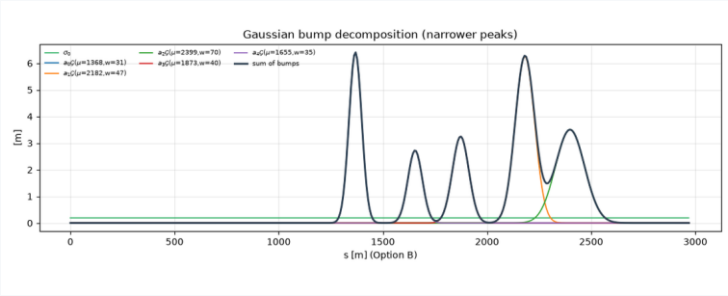
$$\omega_{\mathrm{ref}} = \Delta \mathrm{unwrap}(\psi)/\Delta t; \quad \hat{\sigma} = 1.4826\,\mathrm{MAD}(r);\ \text{use } |v| > 1\ \mathrm{m/s} \text{ for } a_{\mathrm{ref}}.$$

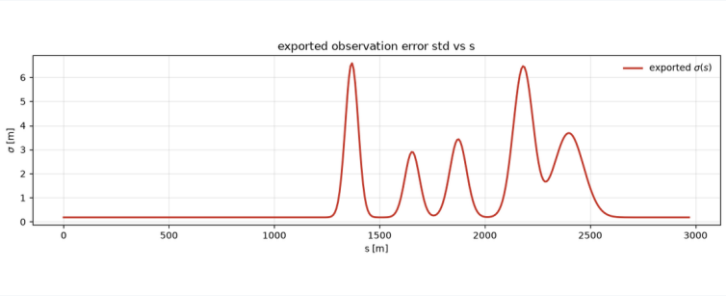
Align stamps; reject bad Δt. For Q: $\sigma_{d_x}^2 = (\sigma_{v_x}\Delta t)^2 + (\tfrac{1}{2}\sigma_{a_x}\Delta t^2)^2$; differencing inflates a noise.

Visual error along the track | Lap3 vs. GT and five-Gaussian fit

Feature detection, lighting, mismatches, and PnP outliers make visual error scene-dependent. Bin-MAD and five peaks fit position-dependent observation noise for EKF covariance R(s).







Lap3 s-order: 1191/1235 valid, median 0.37 m. Peak s≈1.37/1.66/1.87/2.18/2.40 km; a raw spike appears near 2.7 km. Raise R(s) there to downweight vision.

EKF: Motion Prediction + Async Visual Correction

At runtime, SE velocity, acceleration, and yaw rate drive prediction; the first pose only initializes. GT and SE pose series are evaluation baselines.

Inputs and roles

SE • ~106 Hz
Body v / a / yaw rate → Predict

fr / rl • ~10 Hz
Image localization → async observation

Initial SE pose
Initializes x_0, P_0 only

GT / SE pose series
Evaluation only; not an EKF input

Pipeline | Predict → Localize → Time-align → Fuse → Update

Motion input u, Δt
SE control

$\mathbf{x}^-,\ P^-$
motion prediction

$\hat{\mathbf{z}}_c(t_c)$
camera localization

$\hat{\mathbf{z}}_c(t_{\text{last}})$
propagate + fuse

$\mathbf{x}^+,\ P^+$
measurement update

State, motion model, and EKF update

01 Predict

$$\mathbf{x} = [x, y, \psi]^{\mathrm{T}}, \quad \Delta\mathbf{p}_b = \mathbf{v}_b\Delta t + \tfrac{1}{2}\mathbf{a}_b\Delta t^2$$
$$\psi_m = \psi + \tfrac{1}{2}\omega_z\Delta t, \quad \mathbf{p}^- = \mathbf{p} + R(\psi_m)\Delta\mathbf{p}_b$$
$$\psi^- = \mathrm{wrap}(\psi + \omega_z\Delta t)$$

02 Covariance

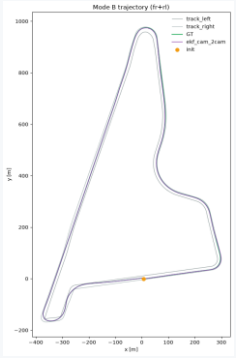
$$F = \partial f/\partial\mathbf{x}, \quad P^- = FPF^{\mathrm{T}} + Q, \quad G = \mathrm{blkdiag}(R(\psi_m), 1)$$
$$Q = G\,\mathrm{diag}(\sigma_{d_x}^2, \sigma_{d_y}^2, (\sigma_{\omega_z}\Delta t)^2)G^{\mathrm{T}}$$
$$\sigma_{d_x}^2 = (\sigma_{v_x}\Delta t)^2 + (\tfrac{1}{2}\sigma_{a_x}\Delta t^2)^2 \ \text{(same for } d_y)$$

03 Correct

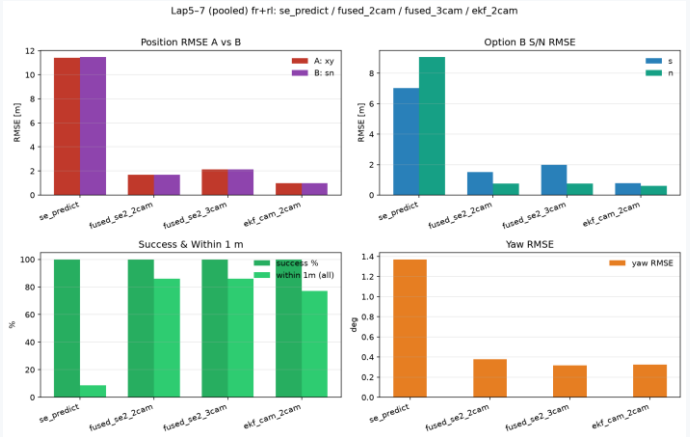
$$\mathbf{r} = \mathbf{z} - \mathbf{x}^-, \quad r_\psi = \mathrm{wrap}(r_\psi), \quad H = I$$
$$K = P^-H^{\mathrm{T}}(HP^-H^{\mathrm{T}} + R(s))^{-1}$$
$$\mathbf{x}^+ = \mathbf{x}^- + K\mathbf{r}, \quad P^+ = (I - KH)P^-$$

$R(s) = \mathrm{diag}(\sigma_{\mathrm{obs}}^2(s), \sigma_{\mathrm{obs}}^2(s), \sigma_\psi^2)$; if vision fails, predict only.

Lap5–7 results | Official workbook + project plots



Lap3 · fr+rl EKF / GT



EKF 2cam xy RMSE 0.976 m

11.7× lower than SE

Method / cameras	xy RMSE (m)	yaw RMSE (°)	Within 1 m	Position median / p95 (m)
SE prediction	11.410	1.370	8.66%	8.058 / 20.953
Offline 2cam fusion (fr+rl)	1.693	0.380	86.01%	0.263 / 2.468
Offline 3cam fusion (r+fr+rl)	2.131	0.316	86.02%	0.288 / 2.456
EKF 2cam (fr+rl)	0.976	0.325	77.16%	0.613 / 2.074

Next Steps: Targeted, Measurable Improvements

Lap5–7 EKF: 0.976 m pooled xy RMSE at 100% coverage. Next: dynamic objects, map quality, lateral constraints, and appearance.

01 Dynamic objects / Mask	**Current**	fr-auto-mask/polygons.json masks the ego car. Other cars may appear as static landmarks and create PnP outliers.	**Next**	Mask detected vehicles before feature extraction; compare mismatches, PnP inliers, success, and error spikes.
02 Map refinement	**Current**	Lap2+4 frames are ordered by track s; Lap3 is held out for validation, avoiding map/query leakage.	**Next**	Optimize landmarks with other training laps via bundle adjustment; keep each test lap held out.
03 Lane lines + 2D track map	**Current**	map.mcap supplies the racing line and track_bounds the edges; the EKF uses x/y/ψ without an explicit lateral n constraint.	**Next**	Use lane/edge detections with track_bounds as soft n observations; check lateral RMSE (Lap5: 0.849 m).
04 Daytime appearance	**Current**	Most training and mapping data is from night runs; exposure and shadows can weaken retrieval and matching.	**Next**	Add daytime maps or domain adaptation; report Top-K hits, PnP success, RMSE, and σobs(s) by time of day.

Plan: mask dynamics → multi-lap BA → lane/track-bound constraint → day/night generalization. Track cross-lap success, RMSE, inliers, and continuity.